Scalable Inversion of Contests with Correlated Performances, Including Softmax and Multinomial Probit

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Abstract

Multinomial probit choice probabilities over n alternatives are Gaussian orthant integrals, and for thirty years econometrics has computed them by simulation, most commonly with the GHK sampler, whose R draws cost $O(n^2R)$ per probability, n-fold that for a full vector, with error $O(R^{-1/2})$. We observe that the choice event is one-dimensional: the probability that alternative i wins is an integral over the winning value, and one shared survival field prices all n alternatives at once. For covariances under which performances are conditionally independent given a few shared variables (factor, block, nested and hierarchical structures, the restrictions applied multinomial probit imposes in practice), one shared field computes all n probabilities, and matrix-free Jacobian–vector products, in $O(nLQ)$ operations over a lattice of L winning values and Q factor quadrature nodes, and inverts observed probabilities back to utilities under a fixed, known covariance (the Jacobian is exact for factor, block and nested structures, and for trees exact within clusters). At $n = 200$ it is two hundred times faster than GHK at 10^4 draws, with error below the comparison's resolution; GHK's measured cost grows superquadratically (exponent 2.15 to 2.8 across the measured range) against the lattice's near-linear growth, so the ratio widens by orders of magnitude. The same two calls price and invert softmax, Thurstone–Mosteller, factor multinomial probit, and a hierarchy of team and sector models; forward pricing is benchmarked to ten million alternatives.

1 Introduction

A decision maker faces n alternatives and chooses the one with the highest utility. Write $U_i = \mu_i + \varepsilon_i$ with $\varepsilon \sim N(0, \Sigma)$; the probability that alternative i is chosen is

$$p_i \;=\; \Pr\{U_i \geq U_j \text{ for all } j\} \;=\; \Pr\{\tilde{U} \geq 0\}, \qquad \tilde{U}_j = U_i - U_j, \tag{1}$$

an $(n-1)$-dimensional Gaussian orthant integral. This is the multinomial probit model. Its appeal has been understood since Thurstone (1927): unlike the logit family it imposes no independence of irrelevant alternatives, because Σ lets alternatives be close substitutes. Its curse has been understood just as long. The integral (1) has no closed form, and estimation needs it, and its derivatives, at every trial value of the parameters.

*`peter.cotton@microprediction.com`. The reference implementation, its parity-locked ports (Python, R, Rust, JavaScript) and the seeded scripts behind every number in this paper live at `github.com/microprediction/winning`; this revision's tables were produced at repository tag `paper-r1`.

The workhorse answer is the GHK simulator of Geweke (1989), Börsch-Supan and Hajivassiliou (1993), Keane (1994) and Hajivassiliou et al. (1996). GHK factors Σ by Cholesky and writes the orthant probability as a nested sequence of one-dimensional truncated normal probabilities, each conditioning on draws for the previous coordinates. Averaging R such importance weights gives an unbiased, smooth estimate of one p_i at cost $O(n^2)$ per draw. The estimator made maximum simulated likelihood practical and it remains the default in textbooks (Train, 2009). Its costs are also standard: error shrinks as $O(R^{-1/2})$; the log of a simulated probability is biased at $O(1/R)$; gradients inherit the simulation noise; and a full probability vector costs n separate runs. A single likelihood contribution needs only the chosen alternative's probability, so the n-fold cost falls not on the individual log-likelihood but on everything else estimation touches: share inversion, aggregate-share and market models, prediction and welfare across the menu, and the derivative of any of these with respect to all n utilities.

This paper takes a different route to the same object, in a min-wins convention fixed here once: performances are times, $X_i = -U_i$, and the lowest time wins. The choice event is one-dimensional. Contestant i wins when its time beats the field, so

$$p_i \;=\; \int f_i(x) \prod_{j \neq i} S_j(x)\, dx, \tag{2}$$

where f_i is the density of X_i and S_j the survival of X_j: integrate over the winning time x, requiring every rival to come in slower. For independent performances Cotton (2021) computed (2) for all n on a lattice by building the field $\sum_j \log S_j$ once and dividing each alternative out, and inverted the map from probabilities back to μ. The present paper extends the field construction to correlated performances by conditioning: whenever Σ makes performances independent given a few shared variables, the field factorizes, and all n probabilities, Jacobian–vector products, and each inversion iteration cost $O(nLQ)$ for a lattice of L points and Q quadrature nodes; the Jacobian itself is a weighted graph Laplacian applied as an operator, never materialized. No per-alternative repetition; error controlled by quadrature, deterministic for the Gauss–Hermite rules and reproducibly randomized for scrambled-Sobol nodes.

Only utility contrasts and a normalized scale are identified in multinomial probit, so the raw covariance is overparameterized, and the contrast covariance may in principle carry unrestricted parameters. Applied work nevertheless imposes structured covariance as a matter of course, for parsimony and for computation (Train, 2009); the grammars here cover the structured families in common use. The table lists them: the first six rows are nested exactly and each is priced and inverted by the same two calls; the last row is a fitted approximation developed in §6.

model	covariance grammar	source
Luce / softmax	independent, minimum-Gumbel base, $D = \pi^2/6$	exact identity
Thurstone–Mosteller	independent, normal base	Cotton (2021)
factor multinomial probit	$VV' + \mathrm{diag}(D)$	§4.1
teams, conferences	blocks: rank-r effect per cluster	§4.2
market plus sectors	nested: blocks $\times$ global factor	§4.2
hierarchies	tree: uniform shared effects	§4.3
dense Σ	fitted grammar + residual factors (approx.)	§6

Scale motivates second. The next table reports measured wall clock for all n probabilities (Rust kernels, one laptop) against extrapolations of GHK's measured cost. The measurements behind the law: the complete-vector GHK timings at $R = 10^3$ are 0.57, 5.66 and 51.3 seconds at $n = 200$, 500 and 1000 (times at $R = 10^4$ scale by ten), a log-log slope of 2.8 over that range, while the small-n table below gives 2.15 from $n = 10$ to 200: the law is superquadratic and steepening. Extrapolations are shown as brackets over the two measured exponents, and every timing is regenerated by the committed `bench.py`. At these budgets GHK's total variation error is near 10^{-2}; the lattice error is near 10^{-8}. The bracketed rows extrapolate far beyond anything GHK is actually run at; that is their point. Nobody prices a million-alternative field with GHK, and the brackets say what the cost law implies about why.

n	lattice, measured	GHK at 10^4 draws, cost-law bracket	multiplier
10^4	0.18 s	20 hours – 3.7 days	4×10^5 – 2×10^6
10^5	2.7 s	0.3 – 6.5 years	4×10^6 – 8×10^7
10^6	29 s	46 – 4,100 years	5×10^7 – 4×10^9

Section 2 places the method among the alternatives. Section 3 gives the independent lattice race; Section 4 extends it across the covariance grammar, factor to blocks to trees, each with its recursion written out. Section 5 gives the Jacobian operator and the inversion. Section 6 reduces dense covariance to the grammar family, approximately, and reports accuracy per named random-correlation ensemble. Section 7 contains the benchmarks.

2 The alternatives

Deterministic quadrature. Genz (1992) transformed the orthant integral to the unit cube and integrated by adaptive and quasi-Monte Carlo rules; the `mvtnorm` implementation is the standard in R. Accuracy is not the issue: in the benchmark below it agrees with the lattice to the reference's own noise. The shape is the issue. It treats one probability at a time, so the all-n vector at $n = 30$ costs three hundred times the lattice, and the gap widens with n.

Frequency simulation. The crude frequency estimator of Lerman and Manski (1981) counts argmax draws. It is unbiased for the whole vector at once but not smooth in parameters, and its per-entry relative error explodes for small p_i. Measured below: at wall clock matched to the lattice it carries twelve to twenty times the total-variation error, records zero counts on tail alternatives at $n = 30$, and at ten times the budget remains a factor of two to four behind.

GHK. Defined above. Its dominance is documentable rather than folklore. Stata's `cmmprobit` manual states that the likelihood evaluator "implements the Geweke–Hajivassiliou–Keane (GHK) algorithm to approximate the multivariate distribution function" (StataCorp, 2023; Gates, 2006), and the command's entire `intmethod` menu (Hammersley, Halton, or pseudo-random point sets) selects only the sequence feeding the same simulator: the market-leading implementation offers no non-GHK method at all. R's `mlogit` and `mvProbit` default to GHK likewise (Croissant, 2020), and the GHK-based `mvprobit` of Cappellari and Jenkins (2003) carried the method to applied Stata work at large. Two properties matter for comparison: it is smooth in parameters, which frequency counting is not, and it prices one alternative per run. Stern (1992) gave a related decomposition with an analytic leading term.

Bayesian data augmentation. Albert and Chib (1993) and McCulloch and Rossi (1994) sidestep the integral entirely by Gibbs sampling the latent utilities; Imai and van Dyk (2005) implement the marginal data-augmentation version in the `MNP` package, and Loaiza-Maya and Nibbering (2022) carry the Bayesian route to large choice sets by variational methods. For Bayesian inference this is a complete and often excellent answer. It is simply answering a different question than the one benchmarked here: the cost moves into the Markov chain, the likelihood surface is never formed, and choice probabilities appear only as posterior functionals, so there is no pricing call to time or to check.

Analytic approximations. Clark (1961) matched moments of pairwise maxima; Mendell and Elston (1974) and Solow (1990) approximate by sequential conditioning on first and second moments. These are fast, sometimes startlingly accurate, and carry no error control, and the benchmark below measures all three properties at once: Mendell–Elston prices the $n = 10$ vector in two milliseconds within 2×10^{-3} of the reference, then drifts to 3×10^{-2} at $n = 30$ with tail probabilities off by a factor of nearly four, and nothing in the output announces the change. The Clark-type recursion behaves alike. The family is current practice, not history: Bhat (2011) builds the MACML estimator on precisely these approximations, and the `Rprobit` package implements it (Bauer, 2024), with its asymptotics examined by Batram and Bauer (2019). Expectation propagation is the modern member of the same family. We benchmark representative sequential moment approximations through the classical members; MACML's composite-likelihood construction around its analytic ingredient is its own estimator and is not itself benchmarked here.

Minimax tilting. Botev (2017) computes single Gaussian rectangle probabilities with bounded relative error deep into the tails, in dimensions up to roughly a thousand. It is the accuracy reference, and we use it as the referee for our own tail claims; it prices one probability per run.

Every smooth, likelihood-grade method above prices one probability at a time; frequency counting obtains the vector but not derivatives. A single likelihood contribution is content with one probability; the tasks that surround it are not. Share inversion, prediction across the menu, welfare, and derivatives with respect to all n utilities each want the whole vector at every parameter value. Sharing the field across alternatives is the algorithmic point of this paper, and it is what makes the n-fold and the derivative costs disappear for those tasks.

3 The lattice race

In the min-wins convention of the introduction, let $X_i = \mu_i + \sigma_i \epsilon_i$ with ϵ_i i.i.d. from a standardized base density (normal, Gumbel, or any density supplied as code), and write f_i, S_i for the density and survival of X_i.

The algorithm evaluates (2) on a lattice $x_1 < \cdots < x_L$:

1. compute $\log S_j(x_\ell)$ and $\log f_j(x_\ell)$ for all j, ℓ;
2. accumulate the field $L(x_\ell) = \sum_j \log S_j(x_\ell)$;
3. for each i, form the integrand $f_i(x_\ell)\, e^{L(x_\ell) - \log S_i(x_\ell)}$, which is contestant i's density times everyone else's survival, with i divided out of the shared field in log space;

4. sum times the lattice spacing, and normalize the vector.

The cost is $O(nL)$: the field is built once, not once per contestant. Log-space arithmetic keeps hopeless contestants finite (their $e^{L-\log S_i}$ underflows gracefully rather than dividing zero by zero), and the exponent is clipped at -745.

The lattice itself is placed on the winner's bulk, not the ability span. A hopeless contestant wins only by running a winner-class time, so only the region where the winning time lives matters. Bisect the conservative winner c.d.f. envelope $G(x) = 1-\prod_j S_j(x)$ to $[G^{-1}(\delta), G^{-1}(1-\delta)]$ and pad two standard deviations, which makes the truncated tail mass negligible against the accuracy floors reported below. Measured: 33 points on this window reach accuracy that a 500-point ability-span lattice cannot, because the span window's own truncation floors near 6×10^{-11}; on fields with many hopeless contestants the bulk window is one hundred times more accurate at equal budget.

Two byproducts are cheap once the field is built. Any specified removal counterfactual, the probability j wins after deleting i, costs one more division out of the field, and any specified photo-finish tie density is one more lattice sum; §5 identifies the latter with the off-diagonal Jacobian. The full $n \times n$ matrices of such queries are $O(n^2)$ outputs and are priced per query, not free.

4 The covariance grammar

Three structures make contestants conditionally independent given a few shared variables, in increasing order of hierarchy: a global factor, factors per cluster, and a tree of uniform shared effects. Each keeps the shared field and the $O(nLQ)$ bill; each is exact, not fitted. The approximate reduction of a dense covariance to this family is a separate matter and has its own section (§6).

4.1 Factor

Let $X = \mu + Vf + \operatorname{diag}(\sqrt{D})\,\epsilon$ with $f \sim N(0, I_k)$ independent of ϵ, so $\Sigma = VV' + \operatorname{diag}(D)$. Conditional on f the performances are independent with means $\mu + Vf$, and

$$p \;=\; \sum_q w_q\, p^{\text{ind}}(\mu + V f_q), \tag{3}$$

a Q-node quadrature over runs of the independent algorithm, cost $O(nLQ)$. In English: average the independent race over the shared luck. Gauss–Hermite nodes serve $k \le 2$; scrambled Sobol serves $k \ge 3$. Conditioning on the factor is not new: Butler and Moffitt (1982) built exactly this quadrature for the one-factor random-effects probit, and the `lpRR`/`slpRR` interfaces of `mvtnorm` integrate over the factor dimensions of $BB' + \operatorname{diag}(D)$ by Monte Carlo. What those constructions price is one alternative per integral. The field is the new part: one pass shares the survival product across all n alternatives, and §7 measures the difference against the per-alternative version of this same quadrature.

One default matters. When $D_i \ll \|V_i\|^2$ the conditional race is nearly deterministic and the integrand over f is nearly a step function; a fixed 15-node rule then silently loses up to five percent of total variation, and the loss is invisible to lattice refinement because it is points-invariant. Randomized invariant testing against a QMC referee found this. The default order now scales with the sharpness $\max_i \|V_i\|/\sqrt{D_i}$, capped by rank.

The same pass returns the own-parameter slopes $\partial p_i/\partial \mu_i$ at no extra cost (negative in this min-wins convention: raising a time mean lowers the win probability), which §5 uses as the Newton preconditioner.

4.2 Blocks and nested

Assign contestant i to cluster $c(i)$ with loading $v_i \in \mathbb{R}^r$ and let

$$X_i = \mu_i + v_i' a_{c(i)} + \sqrt{D_i}\,\epsilon_i, \qquad a_c \text{ i.i.d. } N(0, I_r), \tag{4}$$

a block-diagonal rank-r-plus-diagonal covariance: teams, conferences, sibling products. Two independences do the work. Conditional on its own cluster's effect a contestant is independent of its teammates; and distinct clusters are independent outright. So the joint survival factorizes across clusters,

$$\Pr\{X_j > x \text{ for every } j\} \;=\; \prod_c G_c(x), \qquad G_c(x) \;=\; \mathbb{E}_a \prod_{j\in c} S_j(x \mid a) \;=\; \sum_q w_q\, e^{S_c(x,a_q)}, \tag{5}$$

with $S_c(x,a) = \sum_{j\in c} \log S_j(x \mid a)$ the cluster's conditional log-survival. In English: $G_c(x)$ is the probability that every member of cluster c survives past x, with the cluster's shared luck integrated out.

Contestant i then wins against a cavity field:

$$p_i \;=\; \int h_i(x) \prod_{c\neq c(i)} G_c(x)\, dx, \qquad h_i(x) \;=\; \sum_q w_q\, f_i(x \mid a_q)\, e^{S_{c(i)}(x,a_q) - \log S_i(x|a_q)}. \tag{6}$$

The term h_i couples i's win density to its own teammates within the shared draw of $a_{c(i)}$; the cavity product supplies the rest of the field. The cost is $O(nLQ_a)$ whatever the number of clusters, because the per-cluster factors are accumulated once. Rank two is special: the conditional integrand is smooth and a tensor Gauss–Hermite rule beats scrambled Sobol at equal nodes by a wide measured margin.

Nested. Add one global factor with couplings g_i and a dial $\gamma \in [0,1]$: conditional on the global draw f_q the model is a block race at shifted abilities, so $p = \sum_q w_q\, p_{\text{block}}(\mu + \gamma g f_q)$, a finite mixture of block races. At $\gamma = 0$ the clusters are isolated, at $\gamma = 1$ fully coupled, and the covariance contribution scales as $\gamma^2 g g'$.

4.3 Trees

Let a rooted tree have the leaf clusters at its bottom and internal nodes t above, and give node t a scalar effect $b_t \sim N(0,1)$ applied with uniform strength λ_t to every leaf beneath it:

$$X_i = \mu_i + v_i\, a_{c(i)} + \sum_{t\,\in\,\text{anc}(c(i))} \lambda_t b_t + \sqrt{D_i}\,\epsilon_i. \tag{7}$$

The implied covariance is hierarchical: two contestants share the variance of exactly their common ancestors. The root's effect is common to every contestant, hence choice-irrelevant, and is fixed to zero. The uniform strength is the crucial restriction, because a shared effect that hits every subtree member equally moves the whole subtree's field by a translation of its argument. Messages therefore remain functions of one variable. Per-leaf loadings on internal effects would force two-dimensional message tables.

The algorithm is two passes over the same lattice.

Upward. Each leaf cluster carries its block factor $G_c(x) = \sum_q w_q e^{S_c(x,a_q)}$ from (5). Each internal node, visited deepest first, combines its children under its own effect:

$$G_t(x) \;=\; \sum_q w_q \prod_{c\,\in\,\mathrm{children}(t)} G_c(x+\lambda_t a_q)\,. \tag{8}$$

In English: $G_t(x)$ is the probability that every leaf below t survives past x, with t's effect integrated by quadrature and each child's message shifted because the effect moves that child's whole subtree at once. Shifted evaluations are linear interpolations on the lattice.

Downward. The root's cavity is one. Each node, visited shallowest first, receives the field of everything outside its subtree:

$$R_t(x) \;=\; \Big[\sum_q w_q\, R_{\mathrm{parent}(t)}\big(x-\lambda_{\mathrm{parent}(t)}a_q\big)\Big] \prod_{s\,\in\,\mathrm{siblings}(t)} G_s(x). \tag{9}$$

Contestant i then wins against its own leaf cavity, $p_i = \int h_i(x)\, R_{c(i)}(x)\, dx$ with h_i as in (6). The cost is $O(nLQ)$ plus $O(L\,Q)$ per tree node.

A depth-one tree with zero strengths reproduces the block race to machine precision, which is tested. Against a 2^{22}-path common-random-numbers referee the recursion sits at the simulation noise floor on every resolvable entry at depths one through four (root-mean-square z between 0.75 and 1.1). Entries below the referee's resolution agree with minimax tilting (Botev, 2017) to 0.3 percent relative at probabilities down to 7×10^{-8}, which is within the tilting estimate's own error. Given adequate factor quadrature, tail accuracy is carried by the lattice, and the grammar kernels price tails that simulation does not resolve.

Remark 1 (A connection to financial practice)**.** *Long-only portfolio weights could be considered as race probabilities, and there is a tree-race covariance under which hierarchical risk parity's bisection geometry (López de Prado, 2016) is exactly a race: give internal node t strength $\lambda_t^2 = \rho_t - \rho_{\mathrm{parent}(t)}$ with $\rho_t = \max(1-2h_t^2, 0)$ for merge height h_t, and leaf i idiosyncratic variance $1-\rho_{\mathrm{first}(i)}$; the implied correlation equals the floored cophenetic matrix. The floor is forced, because uniform shared effects cannot represent negative dependence, and omitting it silently inflates other implied correlations above one. The identity turns dendrogram-consistent concentration limits into contest inversions, but portfolio applications are not this paper's subject.*

5 Jacobians, tie densities, inversion

Proposition 1 (Off-diagonals are photo-finish densities)**.** *For the race* (2) *and $j \neq i$,*

$$\frac{\partial p_i}{\partial \mu_j} \;=\; \int f_i(x)\, f_j(x) \prod_{k\neq i,j} S_k(x)\, dx \;\geq\; 0, \tag{10}$$

and each row of the Jacobian sums to zero.

The zero row sum says a common shift moves nothing. The off-diagonal is the density of a dead heat between i and j ahead of the field. The Jacobian is a negative weighted graph Laplacian and is applied as an operator: individual entries and Jacobian–vector products are byproducts of the same field pass (per quadrature node a Gram structure over the lattice), while materializing all n^2 entries is a choice with $\Omega(n^2)$ cost that nothing downstream requires. Three derivative objects should be distinguished: the continuum identity (10); its quadrature evaluation; and the exact derivative of the normalized, adaptive-lattice numerical map, which differs from the first two through the normalization quotient and grid motion. The implementation exposes the frozen-grid form, which matches finite differences of the returned map at every lattice resolution, alongside the continuum form, whose symmetry serves preconditioning. The stable pair factorization is $[f_i e^{L-\log S_i}] \times [f_j/S_j]$: the first factor is bounded, and the hazard f_j/S_j grows linearly in the normal base and exponentially in the Gumbel base, either of which the exponent clip absorbs. The symmetric square-root split overflows where survivals vanish.

The block Jacobian is exact, with the within-cluster term computed under the cavity; the nested Jacobian is the mixture of block Jacobians; the tree Jacobian is exact within a cluster and Gram-approximate across clusters, which suffices for Newton because residuals always use the exact forward map. Feasibility-critical callers fall back to finite differences.

The inversion rests on a convex program.

Theorem 1 (The inverse exists and is unique on contrasts)**.** *Fix a covariance of full support (any member of the grammar with every $D_i > 0$) and let $W(\mu) = \mathbb{E}\min_i(\mu_i + \epsilon_i)$, $\epsilon \sim N(0, \Sigma)$. Then W is concave and differentiable with $\nabla W(\mu) = p(\mu)$, and its Hessian is $-L(\mu)$, the negative of the graph Laplacian whose edge weights are the tie densities* (10)*. Every tie density is strictly positive, so W is strictly concave on the contrast space $\mathbf{1}^\perp$. For any target with $p_i^\star > 0$ and $\sum_i p_i^\star = 1$, the program*

$$\max_{\mu}\ W(\mu) - \langle p^\star, \mu \rangle$$

is constant along $\mathbf{1}$, strictly concave and coercive on $\mathbf{1}^\perp$, and its unique mean-zero maximiser is the inverse: $p(\mu^\star) = p^\star$. If some $p_i^\star = 0$ the supremum is approached only as $\mu_i \to +\infty$; no finite inverse exists, and a floored target recovers a lower bound on that μ_i.

Proof. A minimum of affine functions of μ is concave and expectation preserves this. Ties have measure zero under full support, so the minimiser is almost surely unique and Danskin's theorem gives $\partial W/\partial \mu_i = \Pr\{i \text{ wins}\}$.

The Hessian requires no independence assumption, and we derive it in three steps. Off-diagonal ($j \neq i$): raising μ_j moves only X_j, and the derivative of $p_i = \Pr\{X_i \le X_k\ \forall k\}$ is the probability flux through the boundary $\{X_j = X_i\}$ of the winning region,

$$\frac{\partial p_i}{\partial \mu_j} \;=\; \int \varphi_{ij}(x,x)\ \Pr\{X_k > x,\ k \neq i,j \mid X_i = X_j = x\}\, dx \;\geq\; 0,$$

with φ_{ij} the joint density of (X_i, X_j) under the full correlated law: the photo-finish density of i and j jointly ahead of the field. Diagonal: translation invariance $W(\mu + c\mathbf{1}) = W(\mu) + c$ forces the gradient to sum to one, hence every Hessian row to sum to zero, so $\partial p_i/\partial \mu_i = -\sum_{j \neq i} \partial p_i/\partial \mu_j$ with no further computation. Reduction: when the law is conditionally independent given shared variables—every grammar member—conditioning factorizes the flux integral into (10)'s product form node by node. The Hessian is therefore $-L(\mu)$ with L a weighted graph Laplacian; under full support ($D_i > 0$, so every pairwise difference has positive variance) each edge weight is strictly positive, the tie graph is complete, and $L \succ 0$ on $\mathbf{1}^\perp$: W is strictly concave there.

Coercivity follows from the elementary bound $W(\mu) \le \min_i \mu_i$: for each i, $\min_k(\mu_k + \epsilon_k) \le \mu_i + \epsilon_i$, and $\mathbb{E}\epsilon_i = 0$. For a nonconstant unit contrast d and interior target, $\langle p^\star, d\rangle > \min_i d_i$, because $p^\star$ puts positive mass on a coordinate where d exceeds its minimum; hence

$$W(td) - t\langle p^\star, d\rangle \;\le\; t\big(\min_i d_i - \langle p^\star, d\rangle\big) \;\longrightarrow\; -\infty$$

linearly in t. Constancy along $\mathbf{1}$ is translation invariance again, exactly when the target sums to one. A strictly concave coercive function on $\mathbf{1}^\perp$ attains a unique maximum, where the gradient vanishes: $p(\mu^\star) = p^\star$. □

The program is the Legendre dual of the social surplus function of McFadden (1981), and share inversion as a convex problem is established territory: Berry (1994) inverts market shares in logit-family demand, Galichon and Salanié (2022) identify the general random-utility inversion with optimal transport, and Chiong et al. (2016) compute it by linear programming in the discrete case. What the field representation adds is not the convex program but its derivatives: the Hessian is the tie-density Laplacian of the same field pass, so gauge-fixed Newton runs matrix-free at $O(nLQ)$ per iteration.

Identification lives on the same contrast space. Choice probabilities are invariant to a common utility shock, so the model depends on Σ only through $P\Sigma P$ with $P = I - \mathbf{1}\mathbf{1}^\top/n$: a factor proportional to $\mathbf{1}$, or a tree-root effect applied to every leaf, is choice-irrelevant. The right fitting objective for a dense covariance is therefore the projected residual,

$$\min_{V,\, D\ge 0} \big\| P\,(\Sigma - VV' - \operatorname{diag} D)\,P \big\|_F^2,$$

not a fit of $VV' + \operatorname{diag}(D)$ *to* $P\Sigma P$ (the projection of a rank-plus-diagonal matrix is no longer rank-plus-diagonal, so the second problem manufactures error the first does not have). The reference implementation minimizes the stated objective (`factor_model_projected`: alternate a top-k PSD step for V in the contrast basis with the exact nonnegative diagonal solve, whose normal equations are $(P\circ P)\,d = \operatorname{diag}(P(\Sigma - VV')P)$), and §6's pipeline consumes only projected residuals downstream, so an input already of the form $VV' + \operatorname{diag}(D)$ is returned undistorted. The introduction's identified-class sentence is to be read in this sense: structured covariance is the estimable and computable restriction on $\mathbf{1}^\perp$, not a claim that structure exhausts identification.

The shipped iteration is stated so it can be checked against the code (`abilities_from_race`).

Inversion (all grammars; min-wins). Given a target $p^\star$ with positive entries summing to one, set $t = \log p^\star$ and initialize $\mu^0 = -(t - \bar{t})/2$. Repeat until $\max_i |r_i| < 10^{-8}$:

1. one field pass: $\hat{p} = p(\mu)$, and own-slopes $s_i = \partial p_i/\partial \mu_i$ (factor grammar: the same pass; block/nested/tree: the slopes of the independent race at matched total variances, one extra $O(nL)$ pass);
2. log residual $r_i = \log \hat{p}_i - t_i$; log-domain slope $d_i = \min(s_i/\hat{p}_i,\, -10^{-6})$;
3. damped diagonal Newton step $\mu_i \leftarrow \mu_i - \operatorname{clip}(\alpha\, r_i/d_i, \pm 2)$, then recenter $\mu \leftarrow \mu - \bar{\mu}$.

Damping: $\alpha = 1$ (0.7 at $n = 2$, where the undamped update on the K_2 photo-finish graph is a two-cycle); for the structure grammars $\alpha = 0.7$, halved whenever the residual fails to shrink.

This is a diagonally preconditioned Newton iteration on log probabilities; Theorem 1 makes the target a well-posed problem, and the clip and recentering confine the iterate to the contrast space. Round trips through the shipped front door at $n = 400$ (the committed `bench.py invert`; twenty clusters, tree of depth three): independent 0.04 s, tree 0.37 s, blocks 0.67 s, factor rank-2 1.7 s, nested 3.7 s, each to a maximum log-probability residual below 10^{-8}. Targets below the forward map's resolution are floored, with the one-sided bound direction the theorem states. Ignoring known structure at inversion time is not a small error: inverting block-generated probabilities under an assumed independent model mislocates abilities by up to three tenths of the field's spread (the pinned configuration: $n = 200$, thirty clusters, within-cluster correlation 0.65; maximum mislocation 2.49 against a spread of 8.24, median 0.34).

6 Dense covariance

The fitting procedure, stated so it can be reproduced. Standardize Σ to a correlation matrix $C = \operatorname{diag}(s)^{-1}\Sigma\operatorname{diag}(s)^{-1}$ with $s_i = \sqrt{\Sigma_{ii}}$; after fitting $C \approx VV' + \operatorname{diag}(D)$, the covariance-scale model is restored as $\operatorname{diag}(s)\,(VV' + \operatorname{diag}(D))\operatorname{diag}(s)$, that is, loadings $s \circ V$ and idiosyncratic variances $s^2 D$, so nothing below loses the original scales. (i) Global factors: the certified quotient fit of §5 (`factor_model_projected`), $k = 3$ throughout, returning (V, D_0) that minimize the identified objective. (ii) The working residual is then projected once, $R = P\,(C - VV' - \operatorname{diag} D_0)\,P$: the raw residual still contains the choice-irrelevant common component the quotient fit rightly ignored, and letting later stages chase it manufactures projected error (measurably: an input already of the form $VV' + \operatorname{diag}(D)$ came back with total variation 1.7×10^{-2} under the raw-residual pipeline, and returns at the 2×10^{-4} node-noise floor under the projected one). (iii) Blocks: average-linkage clustering on $\sqrt{(1 - C)/2}$ cut to 20 clusters; for each cluster, the leading eigenvector of the within-cluster block of R with its diagonal zeroed, one rank-one loading per cluster (clusters are disjoint). (iv) Residual promotion: the top $m = 5$ eigencomponents of what remains of R, diagonal zeroed, appended as further factor columns; negative eigenvalues are not promoted; numerically dead columns are dropped. (v) The closing diagonal solves the identified problem's own normal equations, $(P \circ P)\,d = \operatorname{diag}(P(C - V_{\text{all}}V'_{\text{all}})P)$, floored at 10^{-3}. The constants k, the cluster count and m are fixed, not tuned per ensemble; a spectral rule for m was tested and rejected (it saturates its cap for marginal gains and does not repair the failure case below). What results is a factor-plus-blocks race priced in one deterministic call, and the pipeline is the shipped one-call (`fit_covariance`; `race_probabilities` accepts `cov=` directly), not a research script. At $n = 2000$ the call takes five seconds against thirty-six for a 10^6-draw simulation, agrees with it at its noise floor, and prices the 82 percent of tail alternatives for which the simulation records zero wins.

An earlier draft compared a raw-space fit against a fit of $VV' + \operatorname{diag}(D)$ *to* $P\Sigma P$ and found the latter worse on several ensembles; a referee correctly identified that as a comparison between the raw objective and the *wrong* projected objective ($P\Sigma P$ of a rank-plus-diagonal matrix is not rank-plus-diagonal). With the identified objective of §5 and projected residuals throughout, the ensemble study below runs both arms — a raw top-k eigenfit pipeline and the identified pipeline — and the identified arm no longer pays a penalty on in-class truths while retaining its immunity to choice-irrelevant common components.

Accuracy depends on the generating ensemble, so it is reported per named ensemble of the `randomcov` library (one draw per ensemble at $n = 300$, seed fixed, judged on entries a two-million-path referee resolves; the complete stratified table, all fifteen ensembles by probability band with $\frac{1}{2}\|\cdot\|_1$ and $\|\cdot\|_\infty$, is generated by the committed `run_ensembles3.py`). Representative rows, absolute error:

ensemble	$\frac{1}{2}\|p-\hat{p}\|_1$	$\|p-\hat{p}\|_\infty$	med, $p > 10^{-2}$	med, 10^{-3}–10^{-2}	med, $< 10^{-3}$
factor + sparse links	2.8×10^{-3}	7.1×10^{-4}	1.2×10^{-4}	2.8×10^{-5}	6.5×10^{-6}
sparse-precision graphical	4.8×10^{-3}	5.7×10^{-4}	3.2×10^{-4}	5.1×10^{-5}	5.8×10^{-6}
Wishart	1.2×10^{-2}	3.2×10^{-3}	5.5×10^{-4}	1.1×10^{-4}	1.1×10^{-5}
residuals (dense-strong)	6.0×10^{-2}	1.7×10^{-2}	2.3×10^{-3}	6.8×10^{-4}	5.8×10^{-5}
kernel (Matérn/RBF)	3.8×10^{-2}	1.3×10^{-2}	8.8×10^{-3}	4.3×10^{-3}	4.4×10^{-4}

Eleven further ensembles (Marchenko–Pastur and spiked spectra, LKJ, onion, vine, hierarchical, block equicorrelation, AR(1), elliptope walks, and two others) fall between the first and fourth rows. The kernel row is the failure case, and the mechanism is the length scale, not the kernel. A smooth kernel at long length scale has a rapidly decaying spectrum, which is precisely what a low-rank-plus-diagonal fit wants; those draws sit at the easy end of the table. At short length scale the covariance is locality: correlation mass lives on neighbor pairs, the spectrum flattens, and the rank needed to hold it grows with the number of correlation lengths in the domain, which no fixed $k + m$ supplies. Matérn roughness decays polynomially, so at matched length scale it is the harder low-rank target, but it is exactly the family with a sparse-precision escape through the SPDE connection (Lindgren et al., 2011); the squared exponential lacks that route and instead becomes easy again as its spectrum collapses. The right algorithm at short length scale is sequential conditioning with cheap sparse-precision draws, and a production front door should dispatch on the structure it detects.

7 Benchmarks

Against the field. All-n choice probabilities under the rank-2 factor covariance family of Section 2, against a 2^{20}-point QMC reference. Tail is the maximum absolute log-error over reference probabilities in $[10^{-4}, 10^{-3}]$; the $n = 10$ draw has none. Frequency simulation is run at wall clock matched to the lattice and at ten times that budget; "zeros" counts tail alternatives that received no draws.

	$n = 10$		$n = 30$		
method	time	TV	time	TV	tail
lattice race	2.5 ms	1.8×10^{-4}	4.7 ms	8.3×10^{-4}	0.07
Genz, full vector	46 ms	1.8×10^{-4}	1.40 s	8.3×10^{-4}	0.07
frequency, matched time	—	3.8×10^{-3}	—	1.0×10^{-2}	0.45, 2 zeros
frequency, 10× time	—	9.2×10^{-4}	—	2.0×10^{-3}	0.21
Mendell–Elston	2.0 ms	1.8×10^{-3}	18 ms	3.2×10^{-2}	1.30
Clark-type	1.4 ms	1.1×10^{-2}	17 ms	1.6×10^{-2}	1.33

Genz and the lattice agree to the reference's noise and to each other; the 300× cost ratio at $n = 30$ is the one-at-a-time shape, not the integrator. At the paper's headline scale the shape is decisive: one probability at $n = 200$ costs 4.6 s (values agreeing with the lattice to six digits), so the full vector costs fifteen minutes against 26 ms for the lattice, a factor of 3×10^4. The sequential-conditioning family is the only entry whose error grows silently: its tail entries are off by $e^{1.3} \approx 3.7$ with nothing in the output to say so. The race's tail figure is the reference's own noise; lattice self-convergence places its error near 10^{-8}.

Referee agreement. A replay harness maps each benchmark probability to its difference orthant and prices it through the two field-standard R implementations. `mvtnorm`'s Genz–Bretz agrees with the lattice to within its own reported error bound in every family: 3.8×10^{-7} against a bound of 8.1×10^{-7} at $n = 10$, and 2.7×10^{-8} against 5.8×10^{-8} on a case whose smallest probability is 3×10^{-10}. Botev's minimax tilting agrees to its own Monte Carlo noise, which in relative terms is 2.2×10^{-4} at $p \approx 3 \times 10^{-10}$: the tail claim rests on the named accuracy reference, not only on self-convergence. On the independent family an adaptive-quadrature referee reaches machine precision and the lattice matches it to 1.9×10^{-15}. An invariance battery (two-runner closed form, translation, permutation, symmetry) holds to 10^{-16} throughout, except the two-runner comparison at 7×10^{-9}, which is the lattice discretization itself.

Stress boundaries. An adversarial battery pushes each failure axis until something gives. Depth: against pure-relative adaptive quadrature, relative error is 4×10^{-13} at $p \approx 10^{-30}$ and 2×10^{-8} at $p \approx 10^{-50}$, with the first genuine breakdown near $p \approx 10^{-119}$, a laggard twenty-five standard deviations behind. Inversion: the round trip $p \to \mu \to p$ holds 10^{-9} in log even when the target contains a 10^{-10} entry or ties at the 10^{-9} level. The Gumbel base reproduces softmax to 10^{-15} at $n = 1000$; exact duplicates split exactly, and an ϵ perturbation moves the split linearly. The one operative boundary is factor strength. When loadings make implied correlations exceed roughly 0.9, the argmax integrand loses smoothness in factor space and Gauss–Hermite converges slowly in the node count: at loading scale 4 the 25-node rule still sits at 8×10^{-3} total variation. Scrambled-Sobol factor nodes restore reference-level accuracy (4×10^{-4}) at the same call: heavy correlation wants quasi-Monte-Carlo nodes, not deeper polynomial rules. The default now escalates the family automatically past sharpness 3, in both the Python and R implementations (scrambled Sobol and Halton respectively), which agree to 5×10^{-4} on the worst case; the figures above are the pre-escalation diagnosis.

Against GHK. All-n choice probabilities under a rank-2 factor covariance, against a 2^{20}-point QMC reference. TV is a bulk metric and blind to the tails, so the same tail column as above is reported: the maximum absolute log-error over reference probabilities in $[10^{-4}, 10^{-3}]$, which for probabilities this small is the log-odds error, and which the reference's own counting noise floors near 0.1.

	lattice race			GHK, $R = 10^3$			GHK, $R = 10^4$		
n	time	TV	tail	time	TV	tail	time	TV	tail
10	2.3 ms	1.8×10^{-4}	—	1.0 ms	8.4×10^{-3}	—	8 ms	2.1×10^{-3}	—
50	7.0 ms	1.7×10^{-3}	0.11	21 ms	4.2×10^{-2}	1.10	214 ms	1.2×10^{-2}	0.29
200	25 ms	2.4×10^{-3}	0.15	436 ms	3.2×10^{-2}	1.50	4.8 s	1.2×10^{-2}	0.35

The $n = 10$ draw has no reference probability in the tail band. The lattice's tail figures equal the reference's own noise; GHK's are two to ten times larger in log terms even at ten times the draws, which is the tail blindness the TV column cannot show.

The lattice bounds are the reference's own noise; self-convergence places the lattice error near 10^{-8}. GHK error contracts as $O(R^{-1/2})$ and its measured complete-vector cost grows superquadratically, with exponent 2.15 to 2.8 across the measured range. The lattice is deterministic, supplies smooth matrix-free derivatives free of simulation noise (simulated likelihoods can be differentiated, but inherit the noise), and prices the tails. The claim is scoped to the task measured: pricing and inverting complete choice-probability vectors under a known structured covariance — the inner loop that share inversion, counterfactuals, and derivative evaluation repeat — and for that task, under the structured families applied multinomial probit imposes, we consider GHK replaced. Estimation pipelines built around GHK are a larger object than this loop, and §7's estimation paragraphs report where simulation remains competitive at face value. GHK also retains general rectangle probabilities and locality-structured covariance, and Botev (2017) remains the reference for single extreme tail probabilities.

The structure-aware per-alternative comparator. The fairest rival is not GHK but this paper's own conditioning run the literature's way: for each alternative separately, integrate the conditional independent-race integrand over the same Gauss–Hermite factor nodes (the construction of Butler and Moffitt (1982); `mvtnorm`'s `lpRR/slpRR` are its Monte Carlo form). With the same nodes and the same lattice the two computations are algebraically identical, and measure so: at $n = 200$, rank 2, the per-alternative version agrees with the shared field to total variation 3×10^{-15} and costs 601 times as much (22 s against 36 ms; the committed `bench.py bm`). The comparison isolates the paper's contribution exactly: not the conditioning, which is forty years old, but the field shared across alternatives.

Share inversion under sampling noise. The first estimation exercise is deliberately the saturated case. Simulate $T = 50{,}000$ choices from one fixed menu of $n = 30$ alternatives under a known rank-2 factor covariance and estimate the mean-zero utilities by maximum likelihood. With the menu fixed the model is saturated over the simplex interior, so the exact-gradient maximizer is the inversion of empirical shares, $\hat{\mu} = p^{-1}(c/T)$, and the exercise measures how faithfully each method performs that inversion under multinomial noise. Eight replications:

estimator	RMSE over identified alternatives	median fit time
exact gradient MLE	0.0187	4.9 s
MSL, $R = 100$	0.0305	4.4 s
MSL, $R = 1000$	0.0198	40.7 s

At equal fitting time the simulated inversion pays a 63 percent error premium. The $R = 1000$ figure is close to the exact one and the difference is within replication noise at eight replications; the claim is the equal-time column, not the third row.

Covariate estimation. The overidentified case behaves differently, and we report it as we found it. Draw observation-specific design matrices X_t ($n = 10$ alternatives, $d = 3$ covariates, $T = 800$ observations), utilities $\mu_t = X_t\beta$, one choice per observation, covariance known, and estimate β by individual-level maximum likelihood: the exact path uses lattice probabilities with the analytic score (one Jacobian row per observation, a single field pass); the simulated path uses this paper's own Rust GHK with common random numbers and finite-difference gradients. Eight replications, means with standard errors:

estimator	RMSE($\hat{\beta}$)	median fit time
exact gradient MLE	0.0363 ± 0.0066	41.3 s
MSL, $R = 100$	0.0380 ± 0.0073	5.4 s
MSL, $R = 1000$	0.0367 ± 0.0067	39.6 s

All three estimators reach the statistical floor: with three parameters and eight hundred observations the simulation error averages out, and a well-implemented simulator at $R = 100$ is the fastest of the three at this size. The exact path's time is dominated by per-observation Python overhead (the compiled kernel accounts for under a second of the 41); the comparison at $n = 10$ measures plumbing, not method. What the exact likelihood buys in estimation is not this row of the table: it is freedom from the choice of R, gradients without simulation noise, the cost law as n grows, and diagnostics that simulation smooths over. The last point is not hypothetical: on the classic Fishing data, the unrestricted-covariance MNP likelihood evaluated exactly is boundary-seeking (still rising at loading norms near 4000), a ridge that GHK's simulation noise had been regularizing by accident. The behavior is reproducible from the committed estimator (`winning.mnprobit`), which detects the ridge and reports it as a `boundary_` diagnostic rather than returning the interior point simulation would have manufactured.

The Genz–Bretz score. Our quadrature comparator used `pmvnorm`, which exposes no score, with finite-difference gradients (31 vector evaluations per step at $n = 30$, roughly eight times the entire exact-gradient share-inversion fit). The `mvtnorm` manual (v1.4-2) also documents `slpmvnorm` and, for the reduced-rank covariance $BB' + \mathrm{diag}(D)$ of this paper's benchmarks, `lpRR` and `slpRR`: simulated log-likelihoods and score functions that integrate over the K factor dimensions. That interface is itself the conditioning construction of §4.1 with Monte Carlo in place of quadrature and no shared field across alternatives, which makes it the natural structure-aware comparator for a fuller estimation study.

GHK protocol. The GHK figures throughout use this paper's own Rust implementation: per-alternative sequential conditioning on the reference-differenced covariance with a fresh Cholesky factorization per alternative, pseudorandom draws (xoshiro family) with a per-alternative seed offset so that parameter evaluations share common random numbers, no antithetics, no quasi-random sequence, no variable-reordering heuristic, parallelized across alternatives. Cost figures are complete-vector; the large-n columns of the introduction are extrapolations of the measured law and are displayed as such.

Scale. Ten million contestants price in 245 seconds under the block grammar (ten thousand clusters, 257-point lattice, the streaming field pass; the committed `bench.py tenmillion`), and the Rust classic-calibration kernel runs 232 times the speed of the pure-Python implementation. Throughput at this size is a systems statement; the accuracy evidence lives at the scales the referees above can reach.

Parity. One vector file embeds the inputs and outputs of 22 scenarios spanning every claim above. The four language implementations of §8 replay them: twenty at machine precision, the optimizer-mediated scenarios to 10^{-6}.

8 Availability

`pip install winning` is pure Python; `winning[fast]` adds the Rust kernels as one abi3 wheel per platform. A base-R package mirrors the full interface, and a zero-dependency browser port drives a live demonstration at `winning.microprediction.org`.

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